

EE1060: Differential Equations and Transform Techniques

Practice Problem Set - 08

Practice problems in topics covered till: Lecture 30

Note : The practice problem sets are designed for your personal review and are not required for submission or evaluation. These problems aim to reinforce your understanding of the material and help you prepare effectively. Some of the practice problems are sourced from the course references, and it is strongly recommended that you consult these references for additional practice problems. Due credit is also given to the TAs of the course, who have contributed to the creation of this problem set.

1. Consider a system whose impulse response is given by $h(t) = e^{-t}u(t)$. Compute the output of the system for the following inputs:

- (a) $u(t)$
- (b) $tu(t)$
- (c) $\sin(\omega_0 t) u(t)$

2. Consider the kernel $h(t)$ defined as

$$h(t) = \begin{cases} \frac{1}{T} & \text{for } |t| \leq \frac{T}{2} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

How will the following signals be altered when convoluted with this kernel? Compute the output for each case.

- (a) $u(t)$
- (b) $\sin(\omega_0 t) u(t)$

3. Consider the kernel $h(t)$ defined as

$$h(t) = \begin{cases} 1 - \frac{|t|}{T} & \text{for } |t| \leq T \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

How will the following signals be altered when convoluted with this kernel? Compute the output for each case.

- (a) $u(t)$
- (b) $\delta(t)$

4. Compute the convolution of two signals $x(t) = tu(t)$ and $h(t) = u(t - 1)$.
5. Compute the convolution of two signals $x(t) = \sin(\omega_0 t) u(t)$ and $h(t) = u(t - 1)$.
6. Compute the convolution of two signals $x(t) = e^{2t}u(t)$ and $h(t) = e^{-t}u(t)$.
7. Compute the convolution of two signals $x(t) = u(t - 3)$ and $h(t) = u(t - 1)$.

8. Compute the convolution of two signals $x(t) = e^{st}$ and $h(t) = u(t)$ when $s = 3 + j5$.
9. Compute the convolution of two signals $x(t) = e^t u(t)$ and $h(t) = \delta(t - 3)$.
10. Compute the convolution of two signals $x(t) = e^{2t} u(t)$ and $h(t) = \sin(t) u(t)$

Please Note: Some of the problems presented in this document have been adapted from textbooks referenced for this course. These problems are included to help students practice and understand key concepts. The material is used for educational purposes only. We have made every effort to provide proper citations, but if any are inadvertently omitted, it was not intentional. All copyright and trademark rights belong to the original authors and publishers.